

IMO Training – Number Theory

Level 1 Assignment 1 Solutions

$$\begin{aligned}
 1. \quad & \because 121^n - 25^n + 1900^n - (-4)^n \\
 & \equiv 9^n - 9^n + (-4)^n - (-4)^n \pmod{16} \\
 & \equiv 0 \pmod{16}
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } & 121^n - 25^n + 1900^n - (-4)^n \\
 & \equiv (-4)^n - 25^n + 25^n - (-4)^n \pmod{125} \\
 & \equiv 0 \pmod{125}
 \end{aligned}$$

$$\therefore 121^n - 25^n + 1900^n - (-4)^n \equiv 0 \pmod{2000}.$$

2. As there are 7 ending zeros in $34!$, we can tell $b = 0$.

By summing up alternating digits, we have $80 + b \equiv 59 + c + a \pmod{11}$,

i.e. $a + c - d \equiv -1 \pmod{11}$. So $d - c - a = 10$ or -1 .

On the other hand, $139 + a + c + d \equiv 0 \pmod{9}$, $a + c + d \equiv 5 \pmod{9}$.

So $a + c + d = 5, 14, 23$. By considering parity,

$(a + c - d, a + c + d) = (-1, 5)$ or $(-1, 23)$ or $(10, 14)$.

On the other hand, consider $34! \pmod{1001}$, we have $-10c - d + 1 + a \equiv 0 \pmod{1001}$,

i.e. $10c + d - a = 1$.

Solving, $(a, b, c, d) = (2, 0, 0, 3)$.

3. $10^N - N$ has N digits, in which the first $N - 2$ digits are all 9 and the last 2 digits form a two-digit number equal to $100 - N$. Furthermore, sum of the '9's are at most 873. So the sum of digits of the whole number is either 850, 680, 510, 340 or 170.

Then we work case by case:

Sum of digits	Range of $9(N - 2)$	Value of $9(N - 2)$	N	Checking
850	$832 \leq 9(N - 2) \leq 850$	846	96	✓
850	$832 \leq 9(N - 2) \leq 850$	837	95	
680	$662 \leq 9(N - 2) \leq 680$	675	77	✓
680	$662 \leq 9(N - 2) \leq 680$	666	76	
510	$492 \leq 9(N - 2) \leq 510$	504	58	✓
510	$492 \leq 9(N - 2) \leq 510$	495	57	
340	$322 \leq 9(N - 2) \leq 340$	333	39	✓
340	$322 \leq 9(N - 2) \leq 340$	324	38	
170	$152 \leq 9(N - 2) \leq 170$	162	20	✓
170	$152 \leq 9(N - 2) \leq 170$	153	19	

$$\therefore N = 20, 39, 58, 77, 96.$$

4. Consider $a_n \pmod{4}$.

n	$a_n \pmod{4}$	$a_n^2 \pmod{8}$
1	0	0
2	3	1
3	3	1
4	2	4
5	1	1
6	3	1

The sequence $a_n \pmod{4}$ is periodic with period 6. Sum of squares of terms in one period $\equiv 0 \pmod{8}$.

$\therefore$ required remainder $= 0 + 1 + 1 = 2$.

5. For $n^2 \equiv 444 \pmod{1000}$, $n = 38$ is a solution.

For other solutions m , $m^2 - 38^2 \equiv 0 \pmod{1000}$.

$$(m + 38)(m - 38) \equiv 0 \pmod{1000}.$$

Since the difference of the two terms is 76, by considering exponent of 2, both are multiples of 4. On the other hand, as the product is a multiple of 125, exactly one of the two terms is a multiple of 125.

$\therefore m \equiv \pm 38 \pmod{500}$. Indeed, for all $m \equiv \pm 38 \pmod{500}$, $m^2 \equiv 444 \pmod{1000}$.

If $\exists n$ s.t. $n^2 \equiv 4444 \pmod{10000}$, then write $n^2 = 10000m + 4444$, some $m \in \mathbb{Z}$.

Surely n even, and so $(\frac{n}{2})^2 = 2500m + 1111$. With $\frac{n}{2} \in \mathbb{Z}$, $(\frac{n}{2})^2 \equiv 3 \pmod{4}$.

$\therefore$ Such number does not exist.

6. A solution is $183184 = 428^2$ (This line is sufficient to obtain perfect score!).

To find such a solution, we first notice that $n^2 \equiv 1 \pmod{1001}$, with $1 \leq n < 1001$. Then

$$\begin{cases} n^2 \equiv 1 \pmod{7} \\ n^2 \equiv 1 \pmod{11} \\ n^2 \equiv 1 \pmod{13} \end{cases} \iff \begin{cases} n \equiv \pm 1 \pmod{7} \\ n \equiv \pm 1 \pmod{11} \\ n \equiv \pm 1 \pmod{13} \end{cases}.$$

Denote the system $\begin{cases} n \equiv \pm 1 \pmod{7} \\ n \equiv \pm 1 \pmod{11} \\ n \equiv \pm 1 \pmod{13} \end{cases}$ as (a, b, c) , where $a, b, c \in \{0, 1\}$.

Solving the systems, we can see all such possible squares are

$$573^2 = 328329, 846^2 = 715716, 727^2 = 528529 \text{ and } 428^2 = 183184.$$

7. For sequence $\{a_n\}$ defined by $a_n = 4n - 2$, the first four terms satisfy the required property. The first n terms of the sequence satisfy the required property for any $n \geq 5$ too. (Just try.)

Some students suggested the following, which I think is another very reasonable construction:

For general n , write $N = lcm(3, \dots, 2n - 1)$, the LCM of all possible pairwise sums of $\{1, \dots, n\}$.

Then choose the n terms to be $N \cdot 1, N \cdot 2, \dots, N \cdot n$.

8. If $a \neq b$, then WOLOG, let $a > b$. Rewrite the equation as $(a! - 1)(b! - 1) = c! + 1$.
 We can see $c > a > b$. Then take mod $a!$ on both sides, we have $b! \equiv 0 \pmod{a!}$.
 Contradiction.
 $\therefore a = b$, the equation may be rewritten as $c! = (a! - 2)a!$.
 Consider valuation of 2 of both sides. We have $\nu_2(c!) = \nu_2(a! - 2) + \nu_2(a!)$.
 For $a \geq 3$, $\nu_2(c!) = \nu_2(a!) + 1$. So in this case $c = a + 1$ or $c = a + 2$.
 If $c = a + 1$, the equation becomes $a + 3 = a!$. Take mod a on both sides, $3 \equiv 0 \pmod{a}$.
 So $a = 3, b = 3, c = 4$. Checking: $3! \cdot 3! = 3! + 3! + 4! = 36$.
 If $c = a + 2$, the equation becomes $a^2 + 3a + 4 = a!$. Take mod a on both sides, $4 \equiv 0 \pmod{a}$.
 So $a = 4, b = 4, c = 6$, but $4! \cdot 4! = 576$, while $4! + 4! + 6! = 768$. For $a \leq 2$, we can check case by case that there is no solution. $\therefore (a, b, c) = (3, 3, 4)$.
9. By definition, $pqr \mid (pq - 1)(qr - 1)(pr - 1)$, and so $pqr \mid pq + qr + pr - 1$.
 In particular, $pqr \leq pq + qr + pr - 1 < pq + qr + pr$, so $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$.
 WOLOG, $p \leq q \leq r$. We know $p < 3$ because otherwise $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} < 3 \cdot \frac{1}{3} = 1$.
 So as p prime, $p = 2$. Now $\frac{1}{q} + \frac{1}{r} > \frac{1}{2}$. By similar argument, q cannot exceed 5.
 So as q prime, $q = 3$. Hence $r = 5$.
 Checking, $3(5) - 1 = 14, 2(5) - 1 = 9, 2(3) - 1 = 5$, and $2 \mid 14, 3 \mid 9, 5 \mid 5$.
 $\therefore (p, q, r) = (2, 3, 5)$.
 $\therefore$ The only possible value of pqr is 30.

Remark. These 9 problems are extracted from Round 1, British Mathematical Olympiad (BMO). This contest is open for British school students, and serves as the first selection test for the IMO UK Team. Contestants are given 3 hours to solve 6 problems. The top 100 scorers will be invited to Round 2 of BMO. You may look for more problems of similar level by googling 'BMO papers'.